



Integrated active lithium niobate photonic devices

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We report on the fabrication of integrated active lithium niobate (LN) photonic devices such as waveguide amplifiers and tunable lasers using the photolithography-assisted chemo-mechanical etching (PLACE) technique. Specifically, a maximum internal net gain exceeding 20 dB is achieved in the LN waveguide amplifier, and an electro-optically tunable single-frequency laser with an ultra-narrow linewidth of 454.7 Hz is demonstrated in a high-Q LN microdisk. An electrically driven microring laser is demonstrated by the monolithic integration of a diode laser with an LN microring resonator. We also realize a hybrid integration of passive and active LN microdevices using a continuous lithographic processing approach. The integrated active LN photonic devices have a broad range of applications in light-wave communication, precision sensing and quantum information science. © 2023 The Japan Society of Applied Physics

1. Introduction

Photonic integration shows great potential for creating high-performance, low-cost, scalable and reliable devices for communication, sensing, computation and processing.¹⁾ Lithium niobate (LN) has been considered one of the most suitable material platforms for photonic integrated circuits (PICs) due to its outstanding optical properties, including a wide transparency window (0.35 ~ 5 μm), large optical nonlinearity/electro-optic coefficient, and strong acousto-optic effect.²⁾ The revolutionary manufacturing technology for thin film LN (TFLN) wafers has been developed to provide high refractive index contrast, enabling strong optical confinement. In particular, commercially available TFLN wafers combined with the recent breakthroughs in low-loss nanofabrication techniques for LN nanophotonic components have benefited lots of photonic applications.^{2–8)} Single-mode ridge waveguides with propagation loss less than 3 dB m^{-1} and a total length of more than 1.1 m have been demonstrated using the photolithography-assisted chemo-mechanical etching (PLACE) fabrication technique, showing great potential for wafer-scale integration.⁹⁾ To date, almost all building blocks, such as modulators,^{10–14)} frequency converters,^{15–17)} beam splitters,^{18,19)} quantum light sources^{20,21)} and delay lines,⁹⁾ have been constructed on the TFLN platform with unparalleled performances, featuring low-loss propagation, high-speed modulation, high-efficiency frequency conversion, and low-energy consumption.^{4–6,8,16,17,22)} Therefore, the TFLN platform is suitable for large-scale photonic integration.

However, as an indispensable component of PICs, on-chip microlasers and waveguide amplifiers are difficult if not impossible to fabricate on the TFLN platform, since the LN itself does not provide any optical gain. Compared with LN, III–V or II–IV group direct-bandgap semiconductors are more suitable for light generation. Heterogeneously incorporating the semiconductor materials onto the TFLN platform

can provide optical gain through transfer printing, hybrid integration, flip-chip bonding, heterogeneous integration, and providing on-chip light sources in the C-band using an electrical driver.^{23–27)}

An alternative way to provide the optical gain on the TFLN platform can be realized by doping the gain medium in the TFLN. Both on-chip microlasers and amplifiers have been demonstrated on rare-earth ion-doped (REI-doped) TFLN substrates at various wavelength bands (e.g., ~1550 nm and 1030 nm), depending on the intrinsic optical spectra of the doped ions.^{28–48)} Up to now, the doped TFLN microcavities have exhibited loaded Q factors of an order of magnitude of 10^6 using electron-beam lithography-assisted Ar ion milling,⁴⁴⁾ and $\sim 10^7$ using the PLACE fabrication technique.³⁰⁾ The Pockels effect and $\chi^{(2)}$ effect of LN have also been explored for fast wavelength switching and tuning.²⁷⁾ The combination of the narrow linewidth and fast frequency tuning enables many applications ranging from coherent communication and light detection and ranging (LIDAR) to metrology and sensing.

Meanwhile, waveguide amplifiers have also been demonstrated on doped ridge waveguides, providing a high gain ratio of 23 dB.³¹⁾ The active components on the doped TFLN platforms are often pumped using external semiconductor laser diodes. Recently, an electrically driven compact hybrid microring laser on a doped TFLN platform has been demonstrated by butt coupling a commercial pump laser diode chip.³³⁾ Moreover, a monolithically integrated active and passive waveguide array has been demonstrated.⁴⁹⁾ The excellent optical properties of LN combined with low-loss fabrication and novel device designs lead to unparalleled scalability and functionality.

Although the hybrid integration of TFLN photonics is still in its infancy, its rapid development is paving the way for large-scale photonic integration with unprecedented performances and innovative functions. This review article focuses

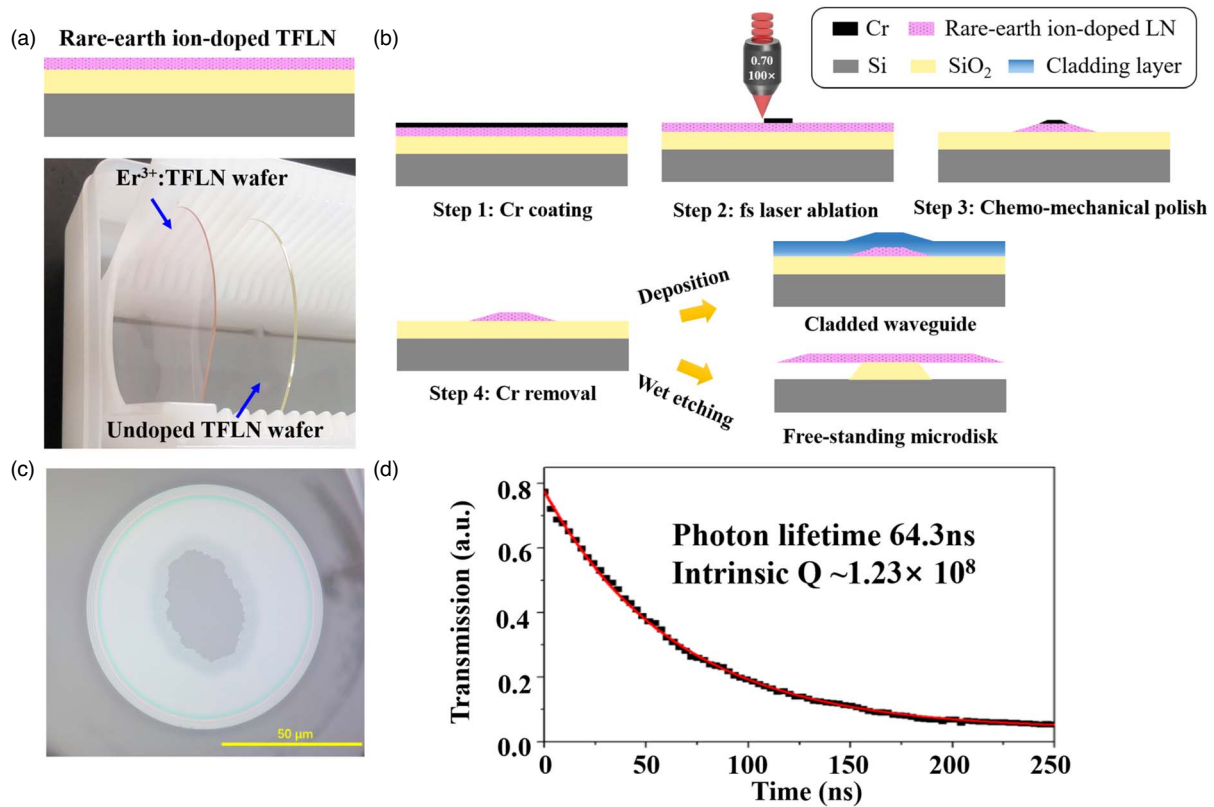


Fig. 1. (Color online) (a) Schematic of the cross-section structure of a rare-earth ion-doped (REI-doped) thin film lithium niobate (TFLN) wafer. Bottom: picture of an Er³⁺-doped TFLN (Er³⁺: TFLN) wafer and an undoped TFLN wafer taken by a digital camera. (b) Schematic of the photolithography-assisted chemo-mechanical etch (PLACE) fabrication process of the REI-doped TFLN photonic device. (c) Overview of the optical micrograph of an ultra high-Q LN microdisk fabricated by the PLACE technique. (d) Transmission spectrum of the microdisk in (c) obtained by ring-down measurement. The intrinsic Q-factor of the microdisk is measured to be 1.23×10^8 .

on the recent development of integrated active TFLN photonics enabled by the PLACE technique. In Sect. 2, we first review the PLACE technique developed for fabricating integrated LN photonic devices. In Sect. 3, we introduce the high-Q LN microresonators and the on-chip microlasers constructed using the PLACE technique. Several applications based on the microlasers are demonstrated. Section 4 describes our latest efforts in fabricating low-loss waveguides and waveguide amplifiers. Lastly, we summarize the aforementioned results and give an outlook on this vibrant and promising field of research.

2. Fabrication technique

The fabrication technique we develop to construct integrated active LN photonic devices is dubbed PLACE.^{50,51)} The technique was proven to be efficient at fabricating ultra low-loss passive LN photonic devices. A record-high intrinsic Q-factor of 1.23×10^8 and an extremely low propagation loss of $0.0034 \text{ dB cm}^{-1}$ were demonstrated in a microdisk and a microring waveguide resonator using this approach, respectively.^{52,53)} The fabrication process mainly includes four steps [as presented in Figs. 1(a) and 1(b)]: first, we coat a chromium (Cr) thin film on top of the LN thin film. Then, we pattern the Cr thin film by femtosecond laser ablation, whilst the LN thin film remains intact without any surface damage because the femtosecond laser intensity is adjusted to be below the damage threshold of LN. The Cr has a much higher hardness than the LN, so the Cr mask can protect the underlying LN in the subsequent mechanical polish.

Therefore, we can transfer the Cr mask pattern to the LN by chemo-mechanical polishing (CMP). Afterward, we remove the Cr metal film using a chemical solution. In addition, a subsequent chemical wet etching and thin film deposition process can be carried out to create free-standing microstructures, cladding layers, microelectrodes and so on. Here, we show a microscope image of an ultra high-Q LN microdisk with a diameter of $100 \mu\text{m}$ [Fig. 1(c)]. The Q-factor is measured by the optical ring-down measurement as shown in Fig. 1(d), and the photon lifetime is about 64.3 ns, which gives an intrinsic Q of 1.23×10^8 .

Thanks to the unique advantage of femtosecond laser direct writing, the footprint of PIC devices is only limited by either the substrate wafer size or the motion range of the positioning stage, both of which can be flexibly scaled up according to specific applications. This technique allows for the expansion of the footprint of PIC devices almost without a physical limit. In addition, the photonic structures fabricated by the CMP process can have a surface roughness below 1 nm, rendering the propagation loss approaching the absorption limit of the substrate material. These unique advantages are critical for developing large-scale or, in the future, even very-large-scale PICs. Figure 2(a) shows the wafer stage of our home-built ultra high-speed high-resolution femtosecond laser lithography fabrication system. A 4-inch TFLN wafer covered with a thin layer of Cr was placed on the moving stage. The femtosecond laser beam was focused on the Cr layer by a high-numerical-aperture objective lens to carry out the ablation process. The Cr

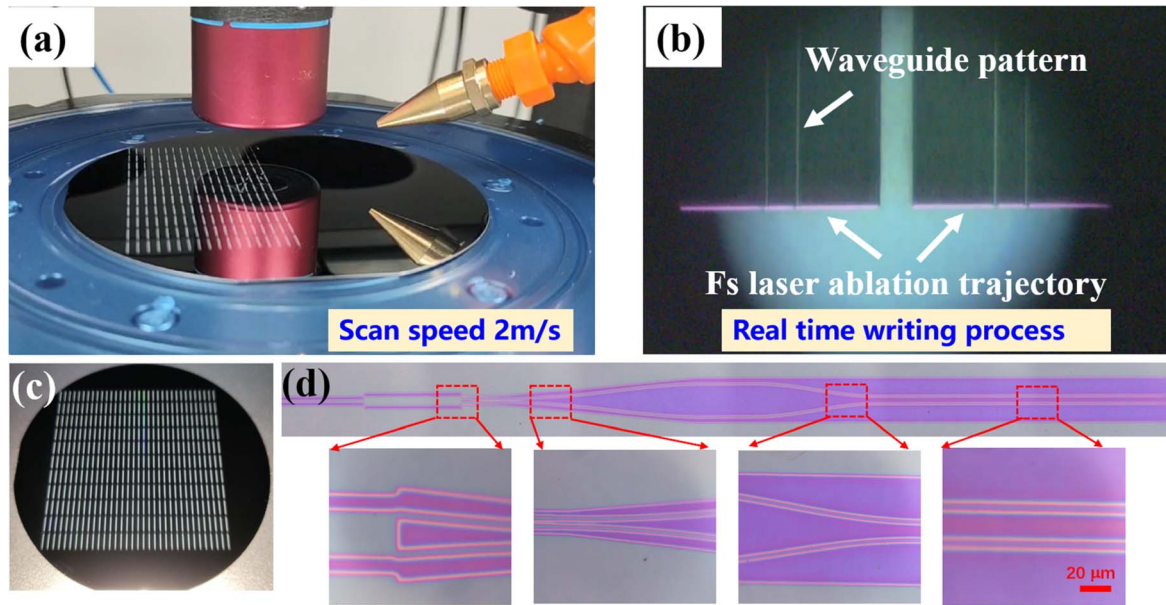


Fig. 2. (Color online) (a) Wafer stage of our home-built ultra high-speed high-resolution femtosecond (fs) laser lithography fabrication system. (b) Microscope image of the real-time femtosecond laser writing process for Cr patterning. (c) A 4-inch wafer with a Cr mask of a 11×68 array of Mach-Zehnder interferometer (MZI) optical modulators. (d) Microscope images of one of the MZIs on the wafer in (c) after the chemo-mechanical polishing (CMP) process.

pattern left on the TFLN formed the mask of an array of 11×68 Mach-Zehnder interferometer (MZI) optical modulators. Figure 2(b) is a microscope image of the real-time writing process of the pattern of two parallel MZIs taken by the CCD camera monitor. The focused laser spot with a diameter of $\sim 0.8 \mu\text{m}$ scanned at a high-speed of 2 m s^{-1} which left behind a pair of pink lines in the picture, indicating a very high manufacturing efficiency. The total time to fabricate the Cr mask for the 748 optical modulators on the 4-inch wafer [as shown in Fig. 2(c)] was only ~ 34 min. This means each optical modulator can be patterned within only 2.7 s. Figure 2(d) is a microscope image of the modulator after the CMP. The bottom insets of Fig. 2(d) show the zoomed-in microscope images of different parts of the MZI structure, indicating a uniform wafer-scale manufacturing quality and a low on-chip optical loss. The optical waveguide fabricated by the PLACE technique shows a trapezoid cross-section and the minimum top width of the optical waveguide is ~ 800 nm.

3. High-Q LN microresonators and microlasers

3.1. Low-threshold on-chip microlaser on an Er^{3+} :TFLN microdisk

Now, we can easily construct high-performance on-chip microlasers by fabricating high-Q microresonators on an REI-doped TFLN platform using the PLACE technique as illustrated in Fig. 1. First, we demonstrate a C-band wavelength-tunable microlaser which is constructed with high-Q erbium (Er)-ion-doped TFLN (Er^{3+} : TFLN) microdisk resonator. As shown in Figs. 3(a) and 3(b), the diameter of the microdisk is $200 \mu\text{m}$ and the Q-factor is determined to be $\sim 1.8 \times 10^6$ at the wavelength of 1563.86 nm by Lorentzian fitting of the transmission spectrum curve. Figure 3(c) displays the transmission spectrum of the microdisk when it is pumped by a diode laser working at a wavelength of around 976 nm . The laser emission is detected near 1563 nm . The upper left inset of Fig. 3(c) shows the close-up view of several laser lines at the C-band

telecommunication wavelengths and the upper right inset of Fig. 3(c) displays the strong green upconversion fluorescence of the microdisk pumped by the 976 nm continuous-wave pump laser. We observe a lasing threshold below $400 \mu\text{W}$ at room temperature, as depicted in Fig. 3(d). The slope of the linear fitting of the power dependence suggests a conversion efficiency of $1.91 \times 10^{-2}\%$.

3.2. Single-mode laser with ultra-narrow linewidth in Er-doped microdisk

Single-frequency ultra-narrow-linewidth on-chip microlasers with fast tunable wavelength play an important role in coherent communication, optical detection, and other applications. According to the Schawlow-Townes theory, the ultra-narrow linewidth is a result of the high-Q-factors, because the fundamental linewidth of lasers is inversely proportional to the quadratic of the loaded Q-factor of the undoped resonator.^{54–56} However, it is still a challenge to fabricate a laser microcavity that has an ultra high optical Q-factor and only supports a single-mode within the optical gain in single microresonators.

To overcome this difficulty, a unique single-frequency ultra-narrow linewidth lasing mechanics was proposed by simultaneous excitation of high-Q polygon modes at both the pump and laser wavelengths in an erbium ion-doped TFLN microdisk.³⁰ These polygon modes are coherently formed by the recombination of the high-Q whispering gallery modes in the weakly perturbed circular microdisk coupled with a tapered fiber. The tapered fiber is used to induce weak perturbation to trigger mode recombination, and couple the light into and out of the microdisk.⁵⁷ Due to the stringent requirement for polygon mode formation, only the single square modes appear within one free spectral range around the output laser wavelength and pump wavelength. Therefore, the polygon modes are sparsely distributed within the optical gain bandwidth compared with the traditional whispering gallery modes. When the pump wavelength is tuned to 968 nm , up-conversion fluorescence and pump light scattered by the microdisk are observed, as shown in the

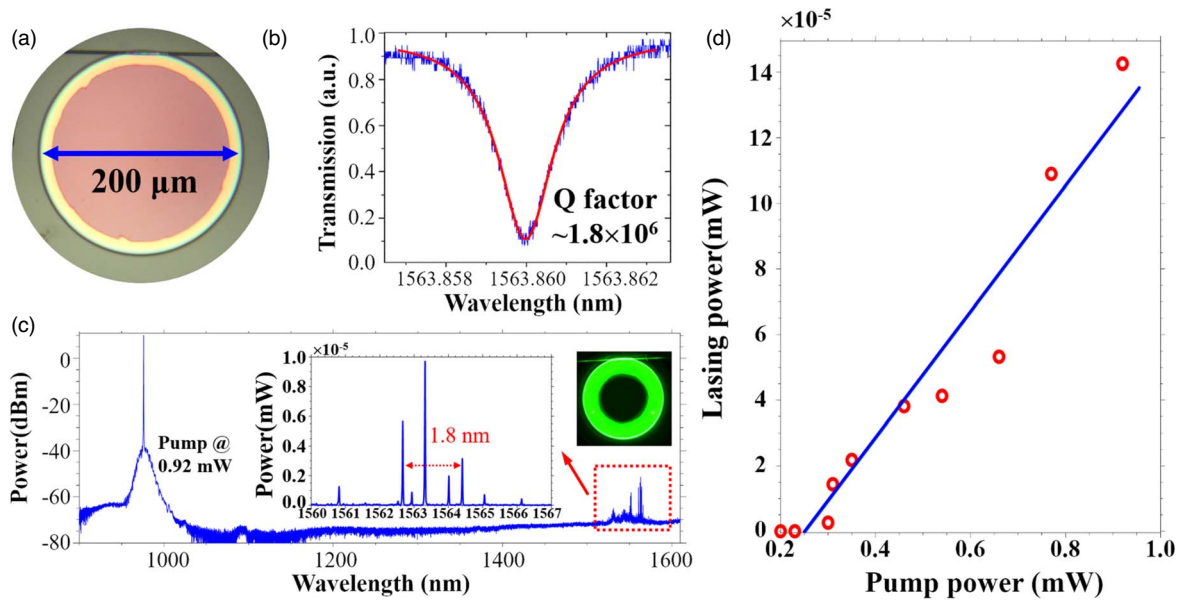


Fig. 3. (Color online) (a) Optical micrograph of an Er^{3+} -doped thin film LN (Er^{3+} :TFLN) microdisk. (b) Lorentzian fitting of the transmission curve indicates the Q-factor of the microdisk reaches $\sim 1.8 \times 10^6$. (c) The spectrum of the Er^{3+} -doped laser with the pump power at 0.92 mW. Upper left inset of (c): enlarged spectrum around 1560 nm marked by a red dashed line block in (c) shows the multimode lasing spectrum. Upper right inset of (c): micrograph of the strong green upconversion fluorescence of the microdisk. (d) Dependence of the intensity of all emission lines on the absorbed pump power at 976 nm, where the experimental data are shown as red circles, and the blue curve is a linear fitting; the lasing threshold is lower than 400 μW .

upper left inset of Fig. 4(a), showing square-shaped trajectories. The spectra of the pump light and up-conversion fluorescence are shown in Fig. 4(a). A single-mode lasing signal is detected at 1546 nm wavelength, and the output power varies with the pump power as shown in Figs. 4(b) and 4(c). The laser mode is also captured, exhibiting a square-shaped pattern as well, as shown in the left inset of Fig. 4(c). The threshold for lasing is determined to be only 25 μW . The theoretically calculated intensity distribution of the lasing mode with a square pattern is also provided in the right inset of Fig. 4(c). The linewidth of the microlaser is measured by

the heterodyne method by beating two independently pumped single-mode microlasers with the same diameter, as shown in Fig. 4(d). The frequency interval of the two lasers (~ 1 MHz) is controlled by adjusting the pump powers. The two laser signals are optically mixed through a 3 dB optical directional coupler and converted into electrical signals through a fast photodetector, as shown in Fig. 4(e). The power spectrum of the beating signal shows a Lorentz-like curve with a full width at half height of 643 Hz. Therefore, assuming that the two microdisks are identical, the linewidth of the microlaser is 454.7 Hz. The frequency

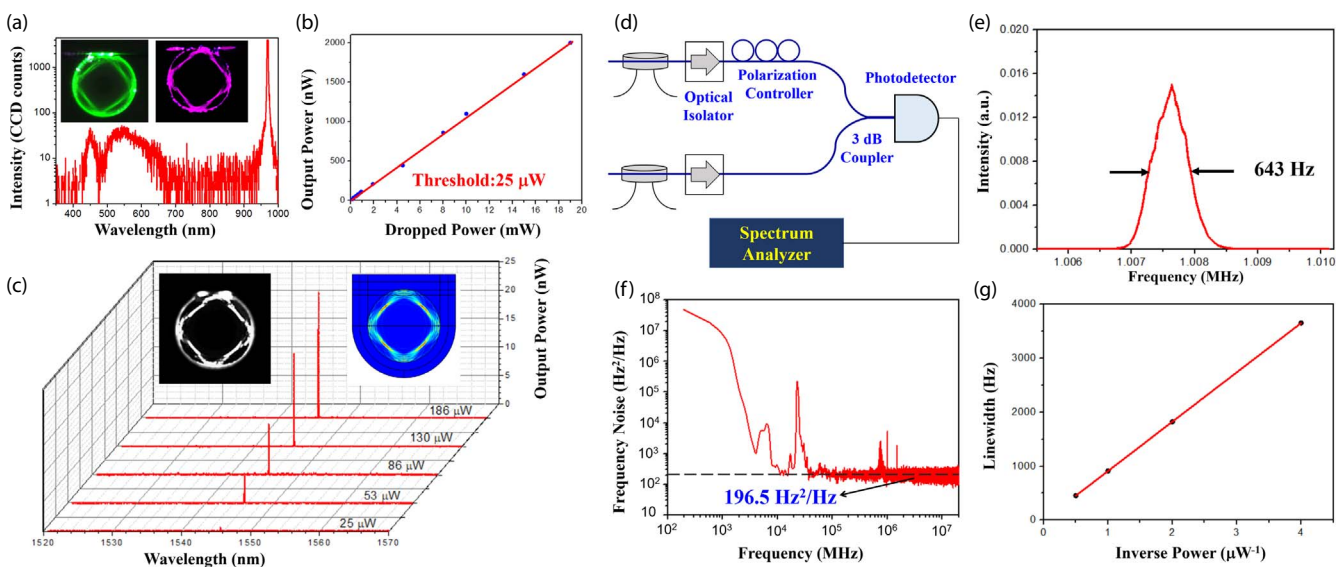


Fig. 4. (Color online) (a) Spectrum of the up-conversion fluorescence and the pump light. Upper insets: optical micrographs of the square modes of the upconversion fluorescence near the 550 nm wavelength (left) and the pump light (right). (b) Laser output power at increasing pump power. (c) Spectra of the microlaser output powers at different pump power levels. Upper left inset: optical micrograph of the lasing mode with a square pattern at 1546 nm wavelength. Upper right inset: theoretically calculated intensity distribution of the square lasing mode. (d) Illustration schematic of the experimental setup for the linewidth measurement. (e) The spectrum of the detected beating signal. (f) Frequency noise is measured to be 196.5 Hz^2/Hz in the high-frequency range, indicating that the fundamental linewidth is 436.6 Hz. (g) Linewidth grows linearly with the output power of the microlaser.

noise is also analyzed, as shown in Fig. 4(f). In the high-frequency range (>0.5 MHz), the frequency noise is almost flat near $196.5 \text{ Hz}^2 \text{ Hz}^{-1}$, indicating that the fundamental linewidth is 436.6 Hz , which agrees well with the result shown in Fig. 4(e). The linewidth of the microlaser as a function of output power is further characterized, as shown in Fig. 4(g). The linearly inverse relationship between the linewidth and the output power agrees well with the Schawlow–Townes formula, indicating that our microlaser linewidth is limited by white noise.

For lots of photonic applications, fast wavelength tuning is usually required. The change of pump power leads to resonant wavelength drift, which provides a method of wavelength tuning, as shown in Figs. 5(a) and 5(b). The wavelength tuning rate is measured to be 12.3 pm mW^{-1} (i.e., $1.54 \text{ MHz } \mu\text{W}^{-1}$). To achieve fast electro-optic tuning, a pair of microelectrodes are fabricated on the top surface of the device. When an external direct current (DC) bias is applied to the electrode pair, electro-optic tuning is achieved. Figures 5(c) and 5(d) show that when the applied voltage is tuned from -300 to 300 V , the laser wavelength exhibits a red shift from 1545.716 nm to 1546.017 nm , indicating a high linear tuning efficiency of $50 \text{ pm}/100 \text{ V}$. The experimental result is roughly consistent with the theoretical calculation value of $80 \text{ pm}/100 \text{ V}$.

3.3. Electro-optically tunable low-phase-noise microwave synthesizer in an active LN microdisk

Very recently, electro-optically tunable dual-wavelength microlasers have also been demonstrated on a single weakly

perturbed Er^{3+} :TFLN microdisk.³⁶⁾ The two laser modes nearly degenerated polygon dual modes with a small wavelength spacing in the same microdisk. A tunable low-phase-noise microwave signal is synthesized by beating the two lasers.

To form the nearly degenerate polygon modes, the microdisk is fabricated with a diameter of $106.33 \text{ } \mu\text{m}$ on a Z-cut Er^{3+} :TFLN wafer. When the wavelength and power of the pump laser coupled into the microdisk with a tapered fiber are tuned to 970.02 nm and above $80 \text{ } \mu\text{W}$ respectively, a dual-wavelength laser signal with a wavelength interval of $\sim 8 \text{ pm}$ is observed, as shown in Fig. 6(a). The up-conversion fluorescence, pump light and lasing light modes emitted by the microdisk are captured and shown in Fig. 6(b). All these optical microscope images exhibit a nearly complete octagonal pattern. Due to the large spatial mode overlap between the pump mode and the two lasing modes, only the nearly degenerate polygon dual modes within the optical gain range of Er ions can be excited. Figure 6(c) shows the laser output power recorded at different pump powers, indicating that the laser threshold is about $80 \text{ } \mu\text{W}$. When the position of the tapered fiber is tuned away from the center, dual-wavelength laser signals with a wavelength interval of 70 pm are observed, as shown in Fig. 6(d). Therefore, the wavelength interval of the dual-wavelength laser signals can be adjusted by the perturbation introduced by the tapered fiber coupling.

To generate the microwave signal, the output power of the two microlasers are first amplified from $1 \text{ } \mu\text{W}$ to 0.047 mW , and the microwave signal is synthesized with a fast

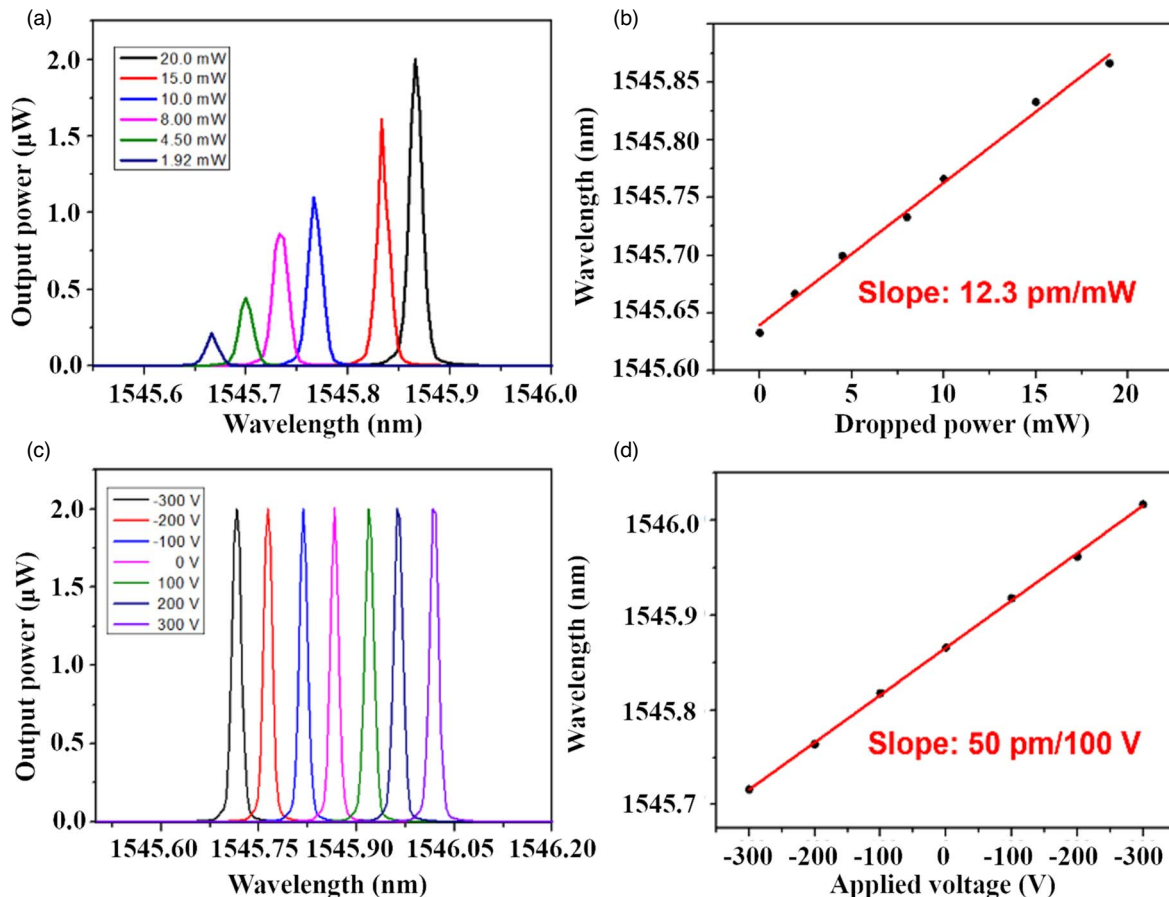


Fig. 5. (Color online) Wavelength tuning performance of the ultra-narrow linewidth microlaser. (a) Lasing wavelength at different pump powers. (b) Lasing wavelength drifts with pump power. (c) Lasing wavelength at different direct current (DC) biases. (d) Lasing wavelength drifts with DC bias.

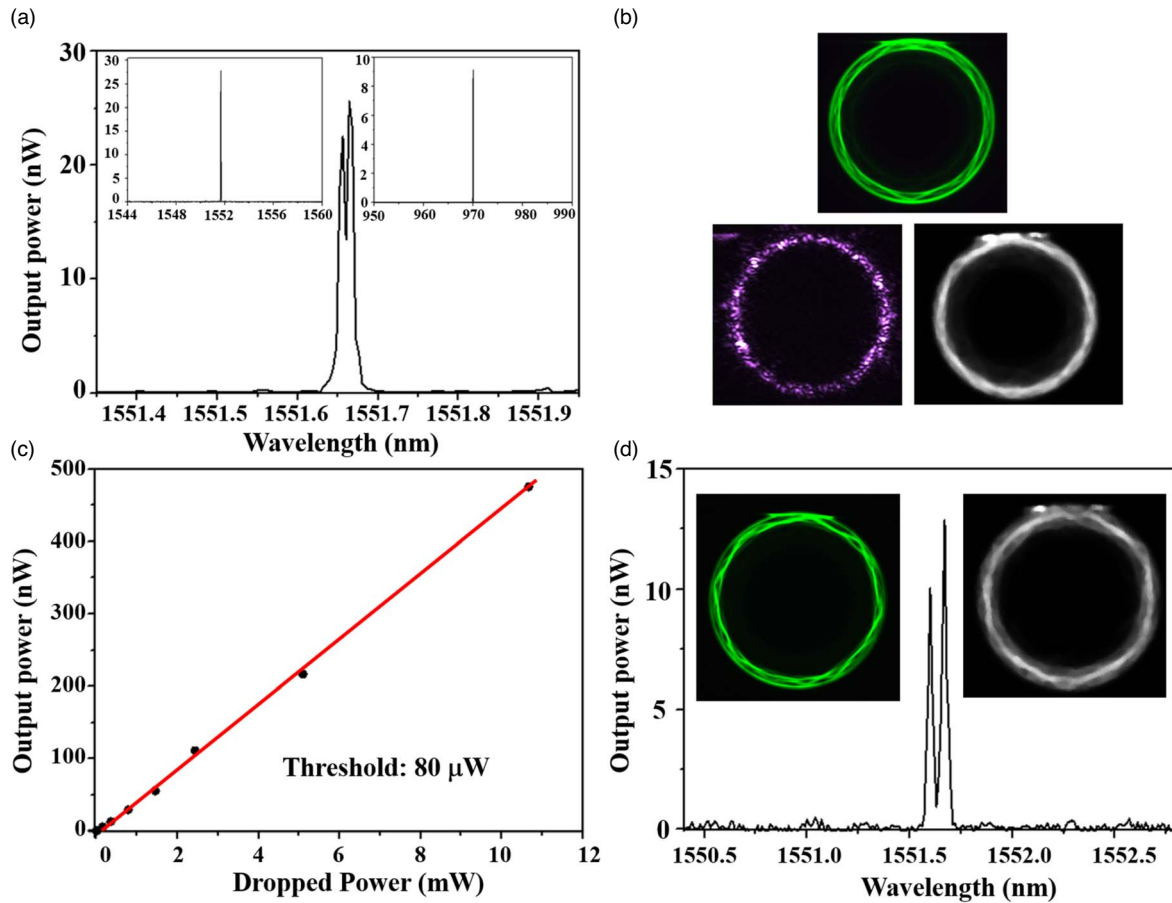


Fig. 6. (Color online) Dual-wavelength microlasers. (a) Spectrum of the microlasers, showing a wavelength interval of 8 pm. (b) Optical micrographs of polygon modes of the up-conversion fluorescence (top), pump laser (bottom left) and output laser (bottom right). (c) Output power of the microlaser as a function of the pump power. (d) Spectrum of the microlasers with a large wavelength interval of 70 pm. Inset: optical micrographs of polygon-shaped patterns of the up-conversion fluorescence (left) and lasing modes (right).

photodetector by signal beating. Figure 7(a) displays the generated microwave signal with a center frequency of 1.23 GHz and a near Lorentz shape. The 3 dB linewidth is measured to be 6870 Hz. Figure 7(b) records the phase noise of the microwave carrier. When the frequency offset exceeds 110 MHz, the phase noise is also flat, which corresponds to the white phase noise of a level of -123 dBc Hz $^{-1}$. Electro-optic tuning of the microwave signal is carried out by applying bias on the paired microelectrodes, as shown in the inset of Fig. 7(a). Figure 7(c) shows the electro-optical tuning result, where the linear electro-optical coefficient r_{22} is used to tune the wavelength spacing of the nearly degenerate dual modes. When the externally applied bias is tuned from -300 V to 300 V, the microwave signal red drifts from 1.73 GHz to 0.73 GHz, indicating a tuning efficiency of -1.66 MHz V $^{-1}$. The dual-wavelength laser with a wavelength interval of 70 pm is also used to synthesize the microwave signal located at 8.72 GHz center frequency, as shown in Fig. 7(d).

3.4. Laser diode-pumped compact hybrid LN microlaser

The PLACE fabrication technology has also been used to fabricate a monolithically integrated TFLN microring laser.³³⁾ Figure 8(a) shows a schematic of the laser diode-pumped compact hybrid LN microring laser by integrating a commercial CoS laser diode with a high-Q Er $^{3+}$:TFLN microring. The on-chip Er $^{3+}$:TFLN microring is fabricated

on a 500 nm thickness Z-cut LN thin film with an Er $^{3+}$ -doping concentration of 1 mol%. Figure 8(b) shows a picture of the fabricated compact hybrid LN microring laser. The diameter of the Er $^{3+}$:TFLN microring is about 1 mm. The microlaser and the monolithic coupling waveguide are clearly visible with the strong upconversion fluorescence. As shown in Fig. 8(c), the input port of the Er $^{3+}$:TFLN microring is physically bonded with the output port of the CoS laser diode to achieve efficient light coupling using a six-axis alignment system. Figure 8(d) shows the spectra of the microring laser with increasing pump power. As depicted in Fig. 8(e), the on-chip lasing power is fitted as a linear function of the on-chip pump power, which indicates a lasing threshold at approximately 6 mW and a conversion efficiency of $3.9 \times 10^{-3}\%$. The lasing performance can be improved by optimizing the coupling efficiency of the pump laser. Finally, we demonstrated that the on-chip laser power is tunable by increasing the driving electric power as shown in Fig. 8(f). The threshold current is measured to be 0.5 A when the operating voltage is set at 1.64 V.

4. Low-loss waveguides and waveguide amplifiers

4.1. Optical true delay line of low-loss waveguide

Figure 9 shows an optical true delay line (OTDL) of a length longer than 1 m. To measure the propagation loss, a 50%:50% beam splitter is fabricated to enable differentiation of propagation losses in two waveguides of very different

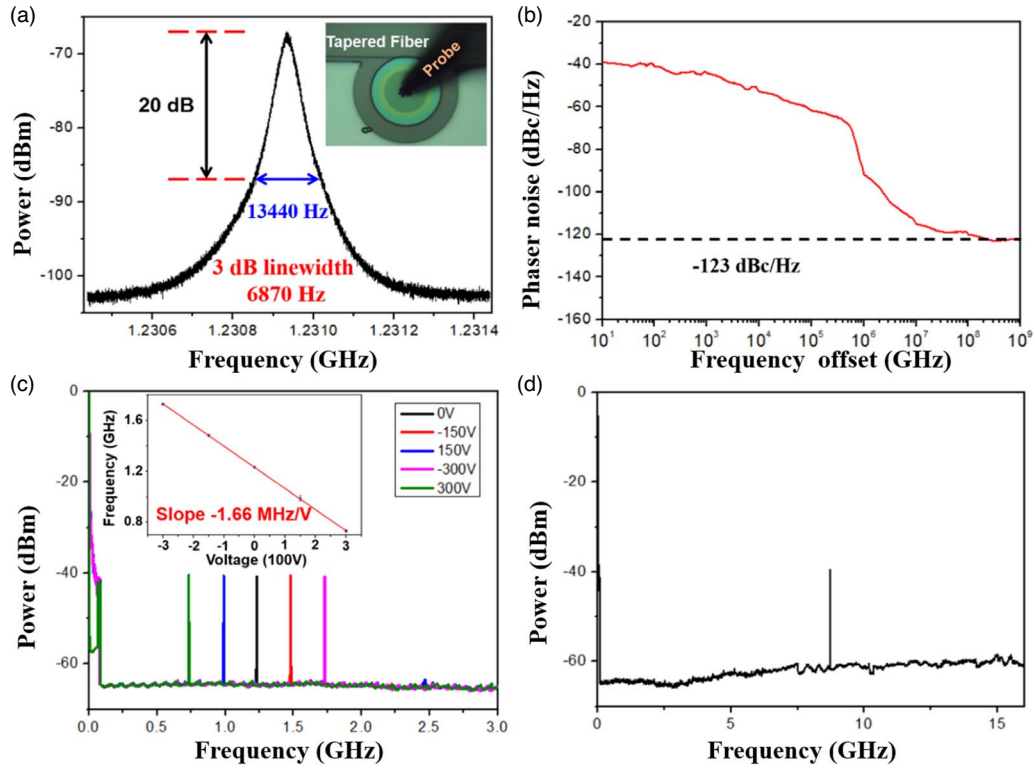


Fig. 7. (Color online) Microwave signal synthesis. (a) Detected microwave signal. Inset: optical micrograph of the probe contracted with the electrode on the center of the microdisk. (b) Phase noise of the microwave signal carrier. (c) Electro-optic tuning of the microwave signal. (d) Microwave signal synthesized by the dual-wavelength laser of a wavelength interval of 70 pm.

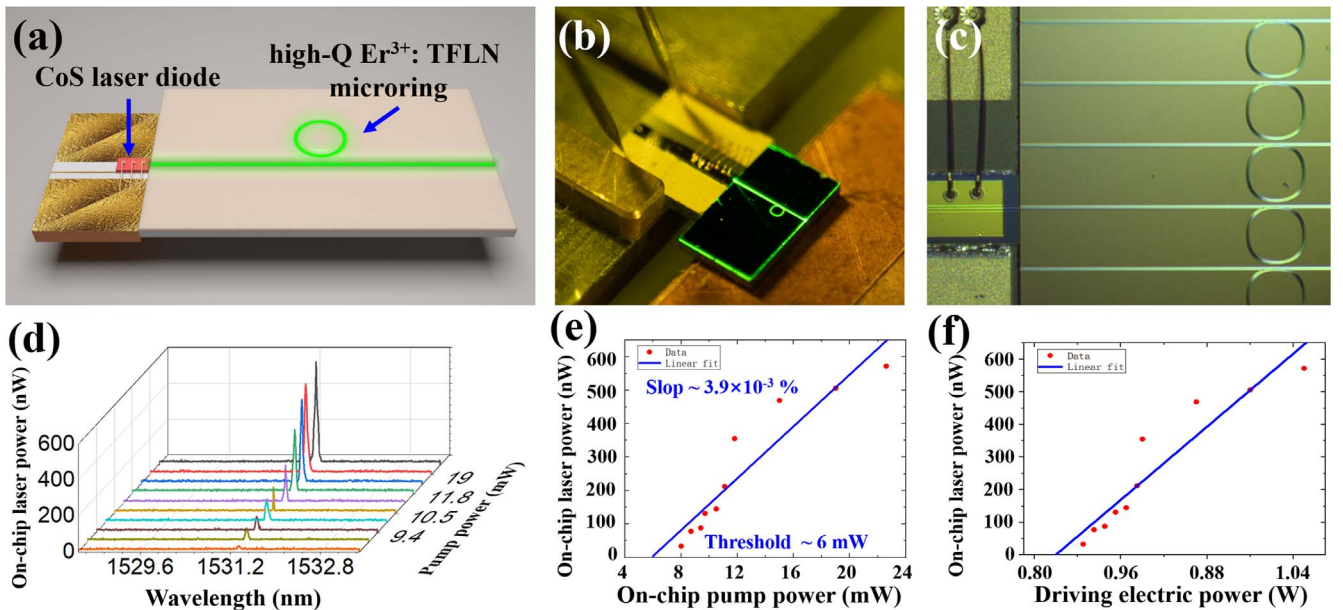


Fig. 8. (Color online) (a) Illustration of the laser diode-pumped compact hybrid LN microring laser. (b) Picture of the microlaser device taken by a digital camera. (c) Top-view optical micrograph of the interface between the CoS laser diode and the array of Er^{3+} :TFLN microrings. (d) Spectra of the microring laser with increasing pump power. (e) On-chip laser power drifts with input pump power. (f) On-chip laser power drifts with driving electric power.

lengths. Figure 9(a) is the schematic design of the beam splitter with a splitting ratio of $\sim 50\%:50\%$.⁹⁾ We fabricate three beam-splitting devices with OTDLs of a long length of 25.06 cm, 54.86 cm, and 111.26 cm after the beam splitter as the upper spiral waveguide shown in Fig. 9(a). The lower waveguide of the three devices has a fixed length of only 2 cm, which is negligible in comparison with the lengths of the upper waveguides. The output powers from the upper and

lower waveguides are measured by the power meter in order to characterize the propagation loss in the OTDLs.

Figure 9(b) shows a picture of the fabricated meter-scale long OTDL taken by a digital camera, which is placed near a 1-Yuan Renminbi (RMB) coin with a diameter of 25 mm. Figure 9(c) shows an SEM image of the area indicated by the blue box in Fig. 9(b). The smooth surface of the waveguides is confirmed by the optical micrograph in Fig. 9(d), in which

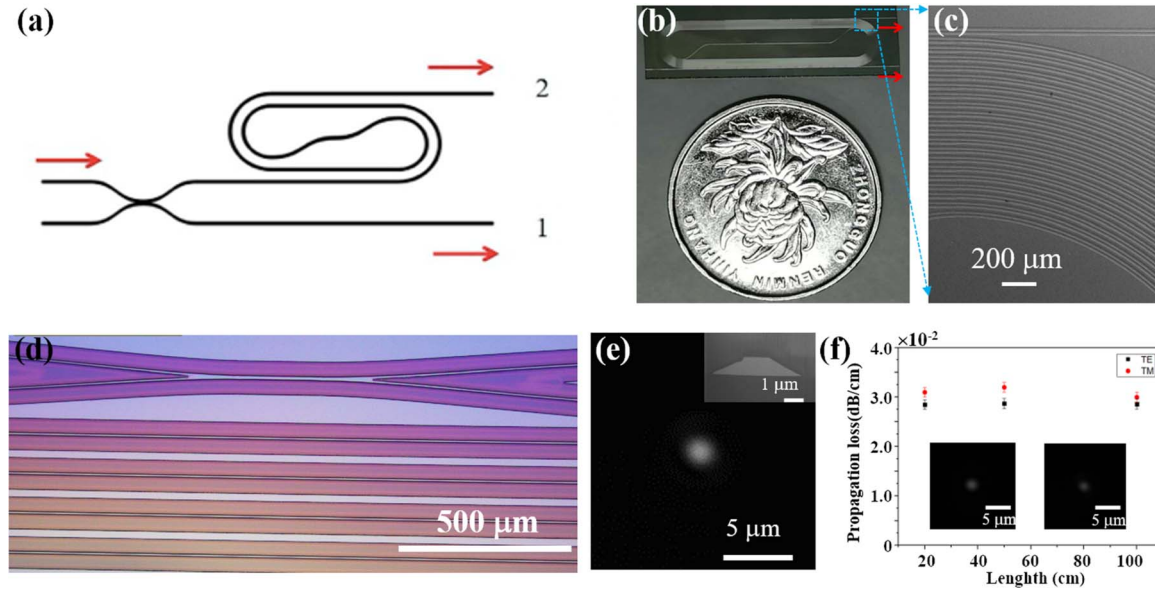


Fig. 9. (Color online) (a) Schematic design of a beam splitter connected with two waveguides of different lengths. The upper waveguide serves as the OTDL. (b) Digital camera picture of an OTDL with a total length of 109.26 cm. (c) SEM image of the bend section of the OTDL in (b). (d) Optical micrograph of the fabricated beam splitter in the OTDL device. (e) Near-field distributions of the modes at the output port 2. Inset: Cross-sectional SEM image of the ridge waveguide. (f) Propagation loss as a function of the length of the OTDL. Inset: near-field distributions of the TE modes at the output ports 1 and 2.

the waveguides and the beam splitter are free from visible defects such as scratches and pits. The beam splitter can be clearly seen with a $130\ \mu\text{m}$ long coupling region. Figure 9(e) shows an SEM image of the cross-section of the waveguide. For the TE-mode, the propagation loss of the OTDL with a length of 109.26 cm is measured to be $0.0286 \pm 0.001\ \text{dB cm}^{-1}$ at 1550 nm wavelength, which is calculated by comparing the output powers from ports 1 and 2. For the TM mode, the measured propagation loss is $0.030 \pm 0.001\ \text{dB cm}^{-1}$ at 1550 nm wavelength.

4.2. EDWA based on TFLN

The technique of fabricating low-loss waveguides can be employed for fabricating the LN waveguide amplifiers presented in Fig. 10. Figure 10(a) shows the schematic of the monolithic Er^{3+} -doped waveguide amplifier (EDWA) chip.²⁹⁾ The 600 nm thick Er^{3+} -doped Z-cut LN waveguide has a top width of $\sim 1.2\ \mu\text{m}$ and a bottom width of $\sim 4\ \mu\text{m}$. The concentration of Er^{3+} ions in the LN waveguides is 1 mol%. Figures 10(b) and 10(c) present zoomed-in images of the curved and straight waveguides of the on-chip LN EDWA, respectively. The spiral design reduces the overall footprint of the amplifier, providing a total gain length of 3.6 cm with a bending radius of $800\ \mu\text{m}$. As shown in Fig. 10(d), the pump light at 980 nm wavelength is provided by a diode laser (CM97-1000-76PM, Wuhan Frelink Optoelectronics Co., Ltd.), while a continuous-wave C-band tunable laser (TLB 6728, New Focus Inc.) with a wavelength range of 1520–1570 nm is used as the signal. The pump and signal laser beams are combined (separated) by the fiber-based wavelength division multiplexers (WDMs) at the input (output) port of the integrated amplifier. A bidirectional pumping scheme is employed to achieve population inversion of the Er ions more uniformly along the full length of the amplifier. Both the output amplified signals and amplified spontaneous emissions are measured by an optical spectrum analyzer. The photograph of a pumped ($\lambda_p = 980\ \text{nm}$) Er^{3+} -doped LN waveguide spiral amplifier chip in Fig. 10(d)

displays strong green upconversion fluorescence along the waveguide. It is observed that the optical signals at the wavelengths of both 1530 nm and 1560 nm experience a rapid increase in gain values along with the growing pump power and a slow gain saturation at the higher pump powers ($>20\ \text{mW}$). Specifically, the internal net gain for the 1530 nm signal approaches the maximum gain of $\sim 18\ \text{dB}$ at the injected pump power of $\sim 40\ \text{mW}$.

4.3. Cladded TFLN waveguide amplifier

The TFLN-based EDWA demonstrated in the previous section utilizes bare TFLN waveguides without a cladding layer which could suffer from fluctuation and instability induced by external perturbations for long-time operation. Thus, it is preferred to deposit a layer of cladding on top of the fabricated TFLN waveguide for direct protection. Besides, the guided optical modes can be manipulated by the cladding layer which allows viable optimization for improved amplifier performance.

Again, we fabricate a 600 nm thick Er^{3+} -doped Z-cut LN thin film prepared by ion-slicing from the bulk Er^{3+} -doped LN crystal, which resides on top of a $2\ \mu\text{m}$ thick silica layer for optical isolation with the underlying 0.5 mm thick Si substrate.³¹⁾ A thin layer of Ta_2O_5 with $\sim 1\ \mu\text{m}$ thickness is deposited as the top-cladding of the TFLN waveguide fabricated by PLACE. A schematic of the waveguide cross-section is shown in Fig. 11(a), showing the LN core with a top width of $2\ \mu\text{m}$ and a bottom width of $6\ \mu\text{m}$. The top-view reflective microscope image and the cross-sectional SEM image of the fabricated TFLN waveguide are shown in Figs. 11(b) and 11(c). The fundamental TE-mode profiles at the pump wavelength (976 nm) and the signal wavelength (1532 nm) employed in the amplifier gain measurement are calculated by the finite element method and depicted in Figs. 11(d) and 11(e) for the air-clad and Ta_2O_5 -clad TFLN waveguides, respectively. The power confinement factors Γ in the LN core for both waveguides are also calculated and denoted in Figs. 11(d) and 11(e).

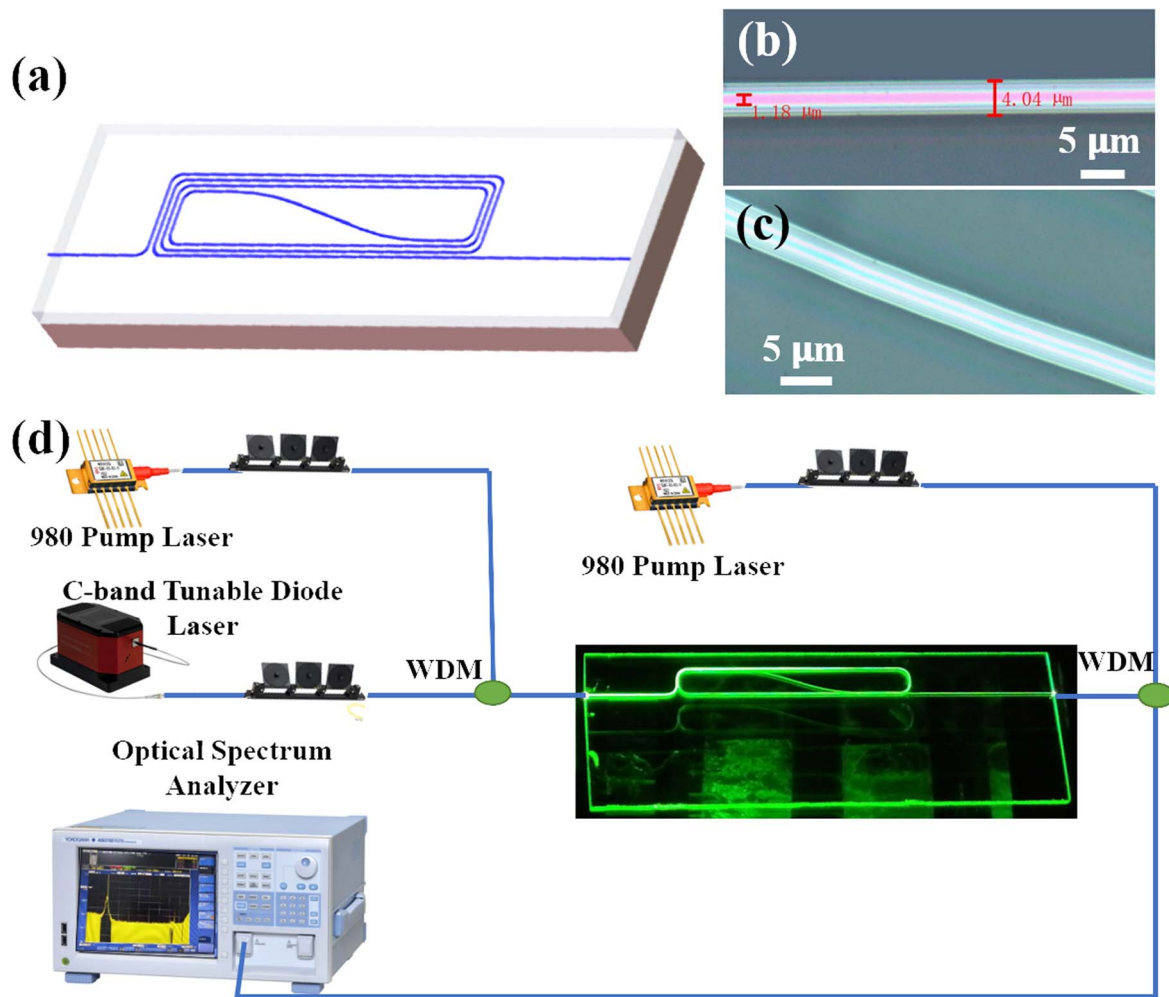


Fig. 10. (Color online) (a) Schematic of the monolithic Er³⁺-doped waveguide amplifier (EDWA). (b) Optical micrograph of the straight waveguide of the on-chip amplifier. (c) Optical micrograph of the curved waveguide of the on-chip amplifier. (d) Schematic of the experimental setup of the gain measurement.

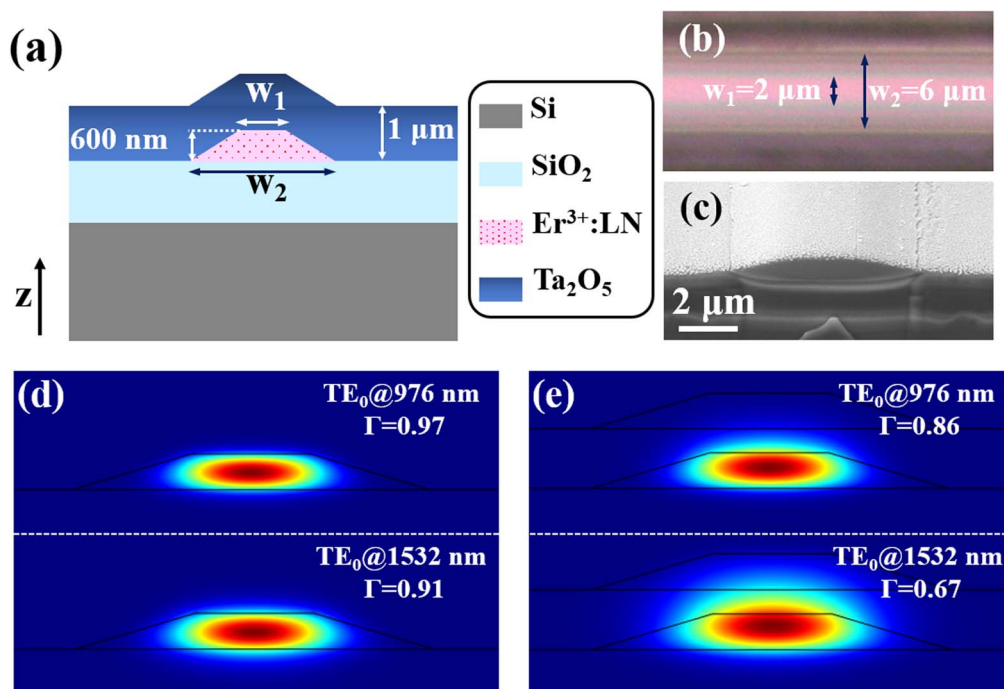


Fig. 11. (Color online) Cladded Er³⁺: TFLN waveguide configuration. (a) Cross-sectional schematic of the Z-cut Er³⁺:TFLN wafer, the fabricated TFLN waveguide, and the Ta₂O₅ cladding on top of the TFLN waveguide. (b) Top-view microscope image of the air-clad TFLN waveguide. (c) SEM image of the Ta₂O₅-clad TFLN waveguide cross-section. (d) and (e) Simulated fundamental TE modes for the air-clad and Ta₂O₅-clad waveguides at the pump and signal wavelengths. The power confinement factor Γ is labelled in each panel.

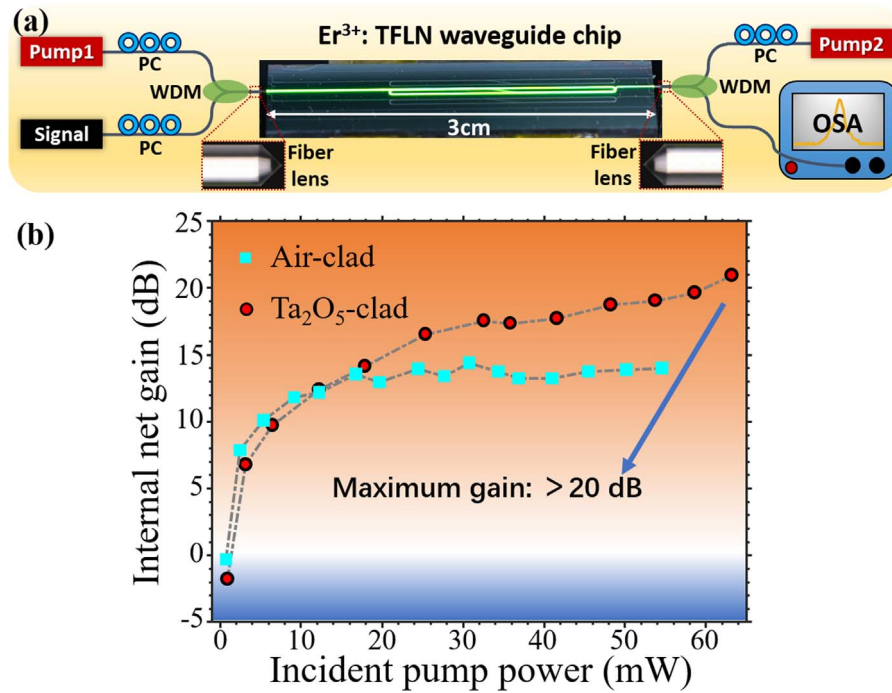


Fig. 12. (Color online) (a) Experimental setup for optical gain measurement. PC, polarization controller. WDM, wavelength division multiplexer. OSA, optical spectrum analyzer. The digital camera photograph of the excited Er³⁺:TFLN waveguide chip is shown in the center. (b) Internal net gain measured from the air-clad (blue squares) and Ta₂O₅-clad (red circles) amplifiers.

To characterize the gain performance of the TFLN waveguide amplifier, ~10 cm long TFLN waveguides with spiral footprints are fabricated and the signal enhancement method is adopted with the experimental schematic shown in Fig. 12(a). The pump and signal waves are combined (separated) by the fiber-based WDMs. The coupling efficiencies of both lensed fibers are measured to be 10 dB (8.4 dB) and 9.4 dB (7.6 dB) per facet for the air-clad and Ta₂O₅-clad waveguides at the pump (signal) wavelength, respectively. All the powers given below are converted into the injected on-chip powers. A digital camera photograph of the excited Ta₂O₅-clad Er³⁺:TFLN waveguide amplifier is also shown in Fig. 12(a) displaying the clearly visible green upconversion fluorescence from Er ions along the full waveguide length.

The measured small signal internal net gain for both waveguides is plotted in Fig. 12(b), where red circles represent gain values from Ta₂O₅-clad waveguide and blue squares denote gain values from air-clad waveguide. It can be clearly seen from Fig. 12 that the internal net gain increases rapidly with increasing pump power and then gradually saturates due to reduced population inversion of Er ions at the high gain regime. It is noteworthy that when the pump power is higher than 12 mW, the net gain of the Ta₂O₅-clad waveguide amplifier is higher than that of the air-clad amplifier, and the maximum net gain we measured from the Ta₂O₅-clad amplifier is above 20 dB at the pump power of 63 mW. The high gain is due to the efficient cladding pump power, as long as the low passive propagation (scattering) loss for the meter-scale TFLN waveguides.

5. Monolithic integration of passive and active devices

The PLACE technique can enable straightforward integration of passive and active TFLN components using a single continuous lithographic mask patterning process, thereby bypassing the troublesome alignment and bonding processes.

Figure 13 depicts the schematic process flow of the robust low-loss optical interconnection for integrated passive and active TFLN photonics.⁴⁹⁾ Firstly, we prepare non-doped and REI-doped TFLN (TFLN: 500 nm TFLN/2 μm SiO₂/500 μm Si substrate) samples [Fig. 13(a)], and a 200 nm thick Cr film is deposited on the top surface of TFLN as the hard mask material for the CMP process. The end facets of the two TFLN samples should be well polished for subsequent seamless stitching. Secondly, we stitch the non-doped and REI-doped TFLN samples tightly using a customized fixture, as shown in Fig. 13(c). In this step, the non-doped and REI-doped TFLN samples of the same thickness are flipped on a polished glass plate for the alignment of the two TFLN samples [Fig. 13(b)]. Ultraviolet (UV) glue is applied on the bottom of the stitched chips and a quartz substrate is used to bond the two samples by irradiation with UV light [Fig. 13(c)], followed by laser welding along the edges of the two TFLN substrates to enable robust bonding upon the quartz substrate [Fig. 13(d)]. Lastly, the photonic circuits are patterned on the stitched LN substrate by femtosecond laser ablation of the Cr thin film, followed by the CMP process which can transfer the waveguide pattern into TFLN [Figs. 13(e) and 13(f)].

The design of the four-channel waveguide amplifier is illustrated in Fig. 14(a), which consists of four Er³⁺-doped spiral waveguides. The four waveguide amplifiers are fabricated on the Er³⁺-doped TFLN while the three multimode interference (MMI) couplers are fabricated on the non-doped TFLN. Figure 14(b) shows an overview of the device fabricated on a TFLN stitch-chip platform. As shown in the false-colored insets of Fig. 14(c), the 1550 nm wavelength signals are in the fundamental mode showing uniform intensity distribution in the four-channel Er³⁺-doped waveguides. The photograph of the amplifier array chip under the optical pumping displays the strong green upconversion fluorescence along the four Er³⁺-doped TFLN spiral

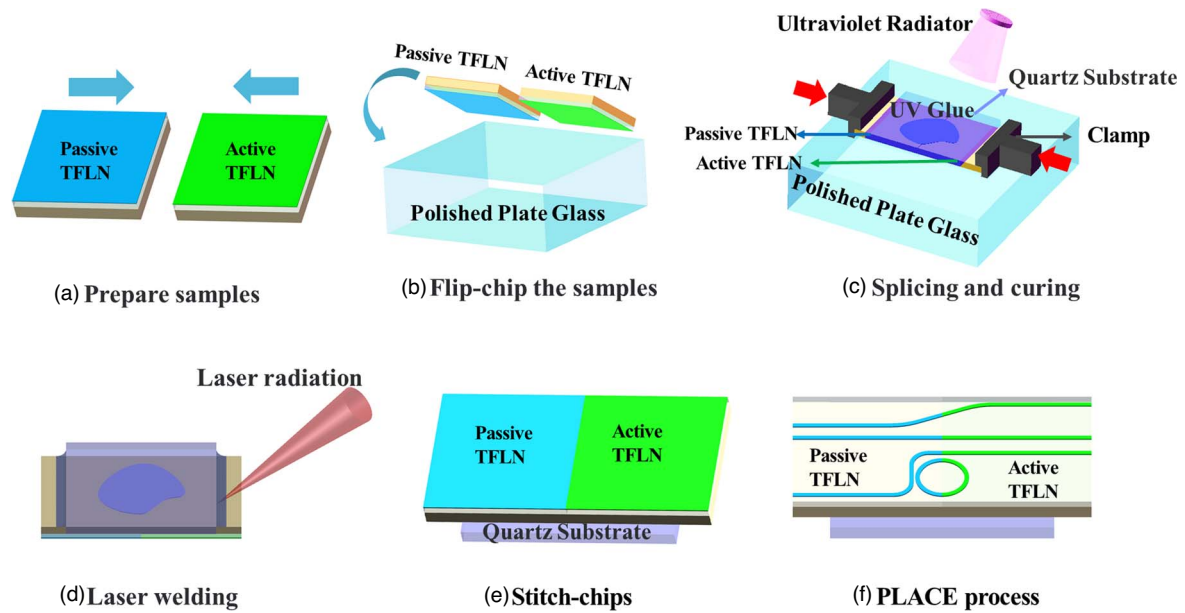


Figure 13. (Color online) Schematic of the fabrication process for robust low-loss optical interconnection of passive and active LN photonics using stitch-chips. (a) Prepare non-doped and REI-doped TFLN samples. (b) Non-doped and REI-doped TFLN samples are flip-chip on a polished plate glass. (c) Non-doped and REI-doped TFLN samples are stitched seamlessly using customized fixtures and the UV glue is applied on the bottom of the stitched chips to bond the two samples. (d) Subsequent laser welding is operated in the boundary of the two TFLN samples and quartz substrate to achieve durable bonding. (e) Completed stitch-chip of passive and active TFLN. (f) Photonic structures fabricated using the PLACE technique on the stitch-chips.

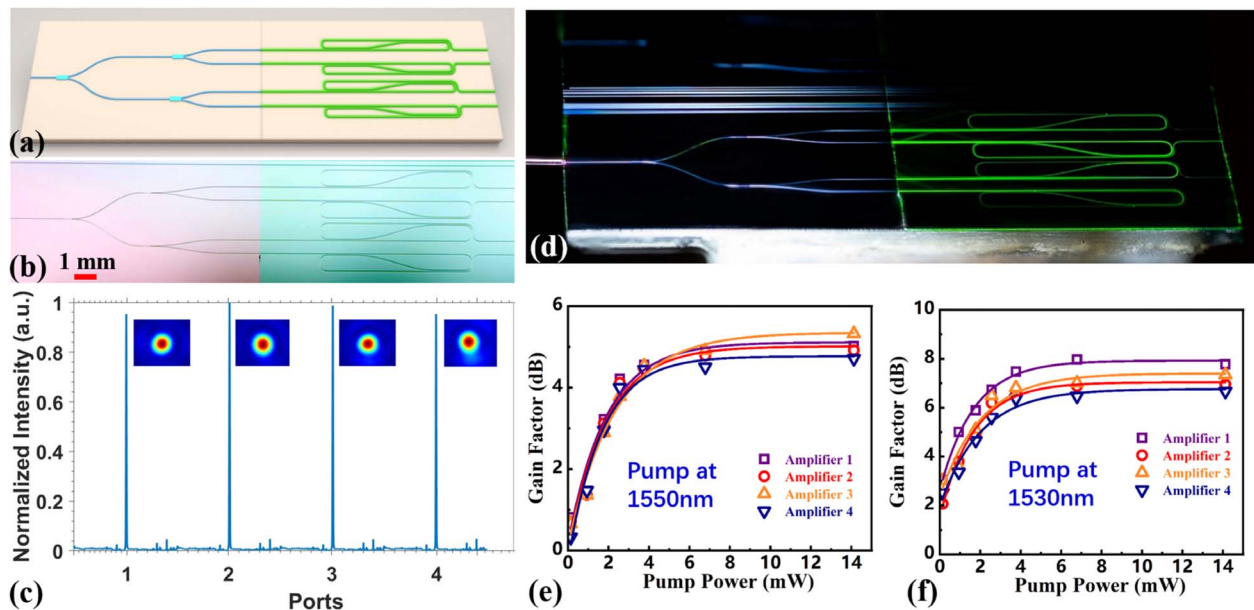


Fig. 14. (Color online) (a) Illustration of four-channel waveguide amplifiers. (b) Photo of four-channel waveguide amplifiers. (c) Mode and intensity distribution of 1550 nm wavelength signal in the four-channel Er^{3+} -doped waveguides. (d) Photo of four-channel waveguide amplifier array when pumped by a 976 nm diode laser. Gain characterization of the four Er^{3+} -doped LN waveguides array for signal wavelengths of (e) 1550 nm and (f) 1530 nm.

waveguides in Fig. 14(d). Figures 14(e) and 14(f) demonstrate the net gain of the integrated amplifier as a function of the launched pump power for the signal wavelengths at 1550 nm and 1530 nm, respectively. The internal net gain for the 1550 nm signal approaches the maximum gain of ~ 5 dB and for the 1530 nm signal approaches the maximum gain of ~ 8 dB.

6. Conclusions and outlook

We demonstrate the fabrication of ultra high-Q microresonators, ultra low-loss meter-scale long waveguides, electro-optically tunable single-frequency ultra-narrow-linewidth microlasers, high gain waveguide amplifiers and integrated

active/passive photonic circuits on TFLN wafers using the PLACE technique. Such integrated active LN photonic devices have a broad range of applications in communication, sensing, computation and processing. Future optimizations in device design, doping quality and fabrication procedures will greatly improve the device performance and pave the way for mass production and widespread applications of integrated active LN photonic devices.

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